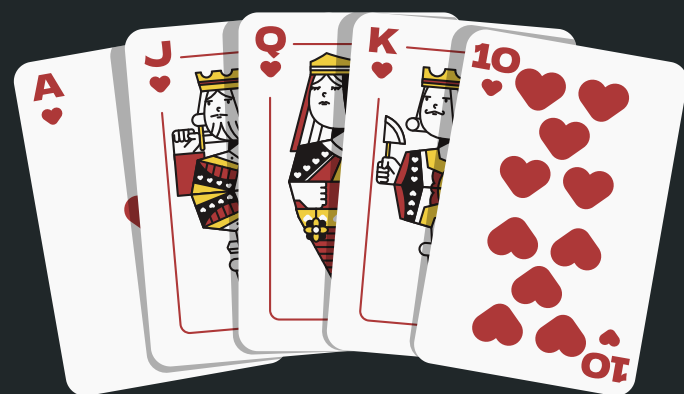
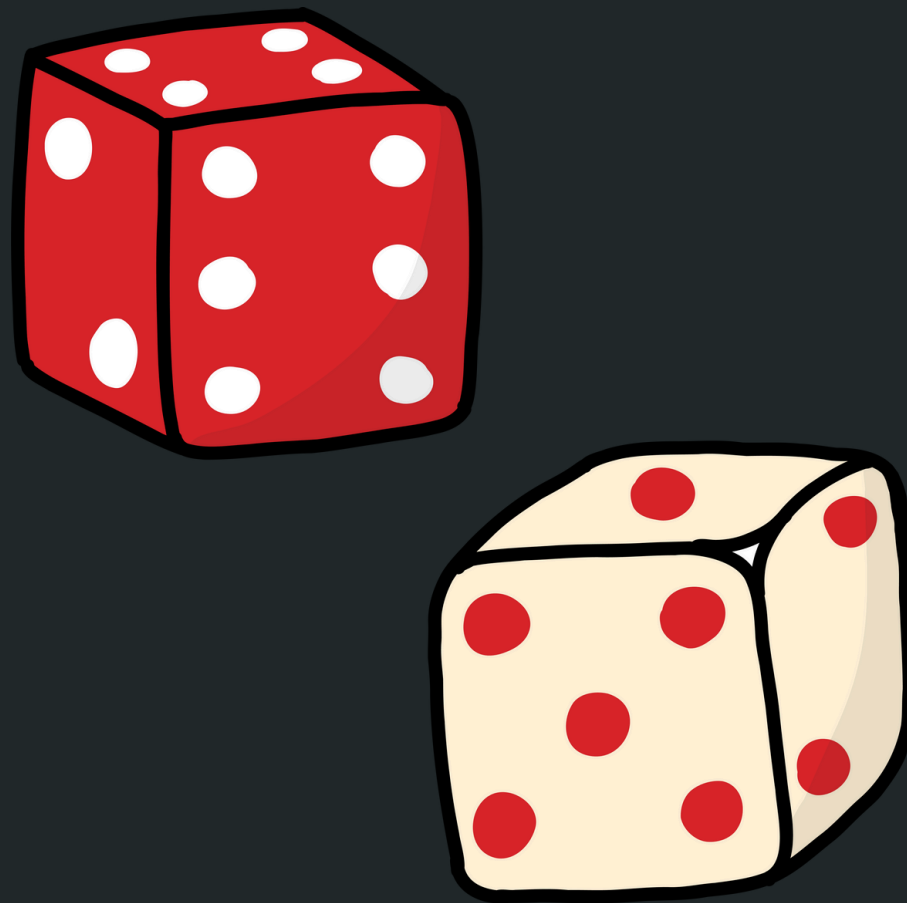


6.12. Probability Basics



What We Learn



Random Events



Probability values



Conditional probability



Independent &
dependent events



Distributions



Use in ML

What Is Probability?

Definition

Probability measures how likely an event is, with values between 0 and 1.

Values always range strictly from 0 to 1.



Random Events - Components



Sample Space

All possible outcomes

e.g: {1, 2, 3, 4, 5, 6}



Event

Desired outcome

e.g: Getting an even number



Probability

Chance of event happening

e.g: $P(\text{Even}) = 3/6 = 0.5$

Simple Probability Examples

Coin Toss



Two equally likely outcomes: Heads or Tails.

$$P(\text{Heads}) = \frac{1}{2} = 0.5$$

$$P(\text{Tails}) = \frac{1}{2} = 0.5$$

Dice Roll



Rolling a specific number is 1 in 6.

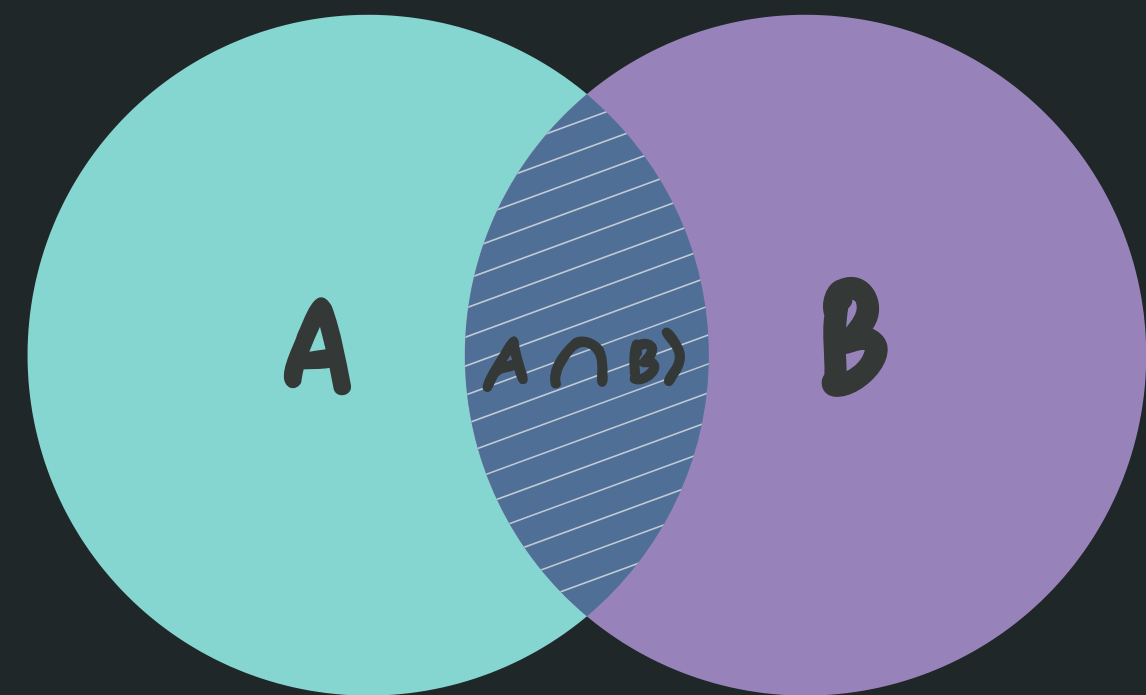
$$P(4) = \frac{1}{6} \approx 0.1667$$

Conditional Probability

Conditional probability answers:
“What is the probability now that I know this extra fact?”

This idea powers models like Naive Bayes, which update predictions using known features.

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$



Conditional Probability - Example



The Scenario

Total Students: 100

Like Machine Learning (ML): 40 students

Among 40 ML-liking students, 25 also like DL.



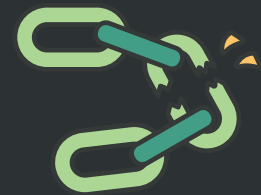
The Calculation

Question: If we pick a student who likes ML, what is the probability they also like DL?

$$P(DL|ML) = 25 / 40 = 0.625$$

62.5% Chance

Independent vs Dependent Events



Independent

Events do not affect each other's outcomes.

Example:

User clicking an ad and weather conditions (no relation).



Dependent

One event influences the probability of the other.

Example:

Drawing an Ace from a deck, keeping it, and drawing another card.

Independent and dependent events matter in ML because they show whether one feature affects another, which impacts how models make predictions.

Probability Distribution

What is a Distribution?

A distribution describes how likely each possible value of a random variable is.

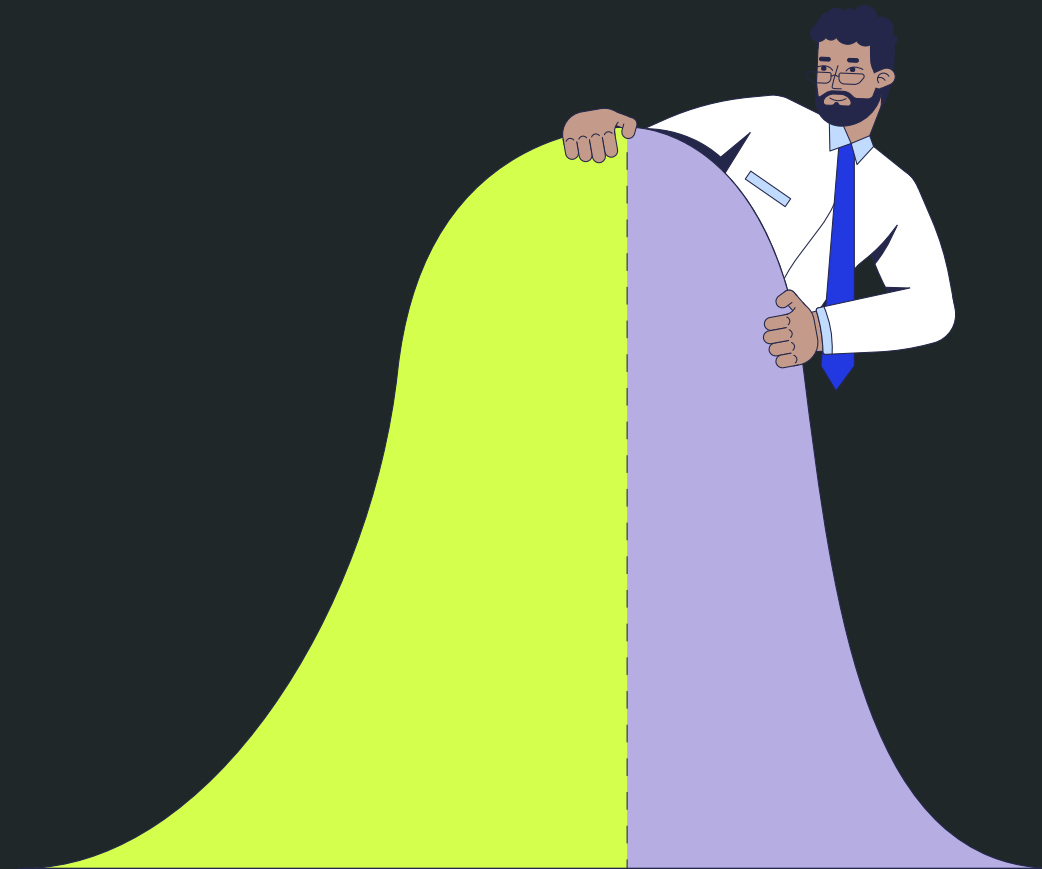
It shows whether values are common, rare, or equally likely.

Importance in ML

In Machine Learning, distributions help us understand data patterns, noise, and uncertainty.

Distribution Types

- **Uniform:** All values equally likely
- **Normal:** Bell-shaped curve (Gaussian)
- **Bernoulli:** One Yes/No outcome
- **Binomial:** Number of Yes/No outcomes across many trials



Applications in ML



Classification

Predicting the probability that an input belongs to a class.



Naive Bayes

A powerful algorithm based entirely on conditional probability.



Sampling

Generating diverse synthetic data examples.